

Program–5: Run Jenkins in a Docker Container

Folder structure:

```
program-5/  
| — Dockerfile  
| — docker-compose.yml
```

STEP 1: Write Dockerfile

nano Dockerfile

```
GNU nano 7.2 Dockerfile  
FROM jenkins/jenkins:lts  
  
USER root  
  
RUN apt-get update && apt-get install -y docker.io  
  
USER jenkins
```

STEP 2: Create docker-compose.yml

nano docker-compose.yml

```
GNU nano 7.2 docker-compose.yml  
version: '3.8'  
  
services:  
  jenkins:  
    build: .  
    container_name: jenkins-ci  
    ports:  
      - "8080:8080"  
      - "50000:50000"  
    volumes:  
      - jenkins_home:/var/jenkins_home  
      - /var/run/docker.sock:/var/run/docker.sock  
  
volumes:  
  jenkins_home:
```

STEP 3: Build & Run Jenkins Using Docker Compose

docker compose up -d

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program5$ docker compose up -d
WARN[0000] /home/1rv24mc120_yashwanth_c/devOps/program5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 209.7s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 547B                                           0.0s
=> [internal] load build definition from Dockerfile                     0.0s
=> => transferring dockerfile: 139B                                     0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts         5.5s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io         0.0s
=> [internal] load dockerignore                                         0.0s
=> => transferring context: 2B                                           0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 150.4s
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.0s
=> => sha256:5687d4d0f728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 65.08MB / 65.08MB 19.3s
=> => sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 5.24MB / 5.24MB 8.2s
=> => sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 3.12kB / 3.12kB 0.0s
=> => sha256:b768bbb699b5f1b6579e23e7253ae1157a339a20a2e9befdf8e03923e0893b800 2.58kB / 2.58kB 0.0s
=> => sha256:75109de20007f9345278f298f4534d443409e234ba81225815caa812cc605644 12.71kB / 12.71kB 0.0s
=> => sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33 49.29MB / 49.29MB 144.8s
=> => sha256:68cdef48009293f6a12f5c50b91fcfe4194bd629334d8a31a45b1ed63cb91a5 1.23kB / 1.23kB 8.8s
=> => sha256:e0e73f34dee93322ae4db1d44258aab1f61011a2b7ec9549ff1f6df342c31168 183B / 183B 9.6s
=> => sha256:68704b042c5d6d57e2ce207c000ee159f2382cb84cb806688c35e7f1ed8627 95.32MB / 95.32MB 86.5s
=> => sha256:501deb283b09e30f43f1f4e27a635bccc1739485e08ba080534552e3d4aae0890 189B / 189B 19.7s
=> => sha256:a9230f7c9c31780c6b1dc4f2aa7e1073dfdf88e9c009f6874e7221ac06245 6.29MB / 6.29MB 22.2s
=> => sha256:d7dd2e3823897146ec4b913a468382968fbd1cc3682e36f56727a1bfd6899 72.05MB / 72.05MB 42.4s
=> => sha256:3fb70446889ebcc58e91f43f90aaf314798941c318ce016ae62068a3de53927 2.23kB / 2.23kB 42.7s
=> => sha256:986be6969b0a04975a2e559b30a96d766bf36d0b340b2789b611688ba8bbea 1.29kB / 1.29kB 43.1s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 392B / 392B 43.4s
=> => extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d612dc14fe368753e7be17b33 1.6s
=> => extracting sha256:5687d4d0f728e3558596e7d718163b2e812a00cf5dd2624459c3315dce34b3cc 1.9s
=> => extracting sha256:2f1ecb09fa362ebe652038ecbfcc64a84b042991e9dedb00059a4127c0121471 0.2s
=> => extracting sha256:68cdef48009293f6a12f5c50b91fcfe4194bd629334d8a31a45b1ed63cb91a5 0.0s
=> => extracting sha256:e0e73f34dee93322ae4db1d44258aab1f61011a2b7ec9549ff1f6df342c31168 0.0s
=> => extracting sha256:68704b042c5d6d57e2ce207c000ee159f2382cb84cb806688c35e7f1ed8627 0.6s
=> => extracting sha256:501deb283b09e30f43f1f4e27a635bccc1739485e08ba080534552e3d4aae0890 0.0s
=> => extracting sha256:a9230f7c9c31780c6b1dc4f2aa7e1073dfdf88e9c009f6874e7221ac06245 0.0s
=> => extracting sha256:d7dd2e3823897146ec4b913a468382968fbd1cc3682e36f56727a1bfd6899 0.6s
=> => extracting sha256:3fb70446889ebcc58e91f43f90aaf314798941c318ce016ae62068a3de53927 0.0s
=> => extracting sha256:986be6969b0a04975a2e559b30a96d766bf36d0b340b2789b611688ba8bbea 0.0s
=> => extracting sha256:d2519f76d4c6527e03084f2742da3ae15b0c5097d464be4dd93de4f573749313 0.0s
=> [2/2] RUN apt-get update && apt-get install -y docker.io          50.9s
=> exporting to image                                                2.8s
=> => exporting layers                                              2.7s
=> => writing image sha256:1cb62d3a7be9cf4dec7a7ff89765f9a1cca8aba013298c7531e2512c0fe2d95 0.0s
=> => naming to docker.io/library/program5-jenkins                  0.0s
```

Check:

docker ps

```
1rv24mc120_yashwanth_c@yashwanth-c-Inspiron-15-3525:~/devOps/program5$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
13780b751cf8   program5-jenkins  "/usr/bin/tini -- /u..." 30 seconds ago Up 30 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
d44b7306e7de   program-2       "docker-entrypoint.s..." 3 hours ago   Up 3 hours   0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
```

STEP 4: Access Jenkins Web UI

Open browser:

<http://localhost:8080>

STEP 7: Get Jenkins Initial Admin Password

```
docker exec jenkins-ci cat /var/jenkins_home/secrets/initialAdminPassword
```

- Copy the password
- Paste it in browser
- Click Continue

STEP 8: Complete Jenkins Setup

Install suggested plugins

STEP 9: Verify CI Setup

Dashboard loads successfully → CI environment is ready.

